## Convergence and Taylor Series — Inconclusive Tests and Omitted Endpoints

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Two errors account for most lost points on series questions, and both are errors of *logic* rather than of algebra.

- **Inconclusive test.** The $n$th-term test can prove divergence and never convergence. A ratio-test limit of 1 decides nothing at all.
- **Omitted endpoint.** The ratio test gives the radius. The two endpoints are outside its reach and must each be tested on their own, by substituting and running a different test.

For every problem, name the test you used and say exactly what it licenses.

**1.** Warm up. A series has terms $a_n = \dfrac{4n+1}{n+7}$. Compute $\lim_{n\to\infty} a_n$, then state what the $n$th-term test lets you conclude about $\sum a_n$.

Answer: ____________________

**2.** **Find and fix.** Devi writes: "For $a_n = \dfrac{1}{\sqrt{n}}$ we have $\lim_{n\to\infty} a_n = 0$, so $\sum a_n$ converges." Devi's limit is correct; her conclusion is not. Compute the limit yourself, state precisely what the $n$th-term test does and does not license here, and name the test that actually settles this series — together with its verdict.

Answer: ____________________

**3. Find and fix.** A student applies the ratio test to $\sum_{n=0}^{\infty} 2\left(\frac{1}{3}\right)^n$, finds the limit is less than 1, and correctly concludes convergence. Asked for the sum, the student writes 6. Identify the formula the student misremembered, then compute the exact sum of the series.

Answer: ________________

**4.** Consider $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$. The ratio test on this series returns a limit of exactly 1, so it settles nothing. Use partial fractions and a telescoping partial sum instead. A classmate reports the sum as 0.5. Compute the exact sum of the series.

Answer: ________________

**5.** Two different series both return a ratio-test limit of 1: $\sum \frac{1}{n}$, whose ratio is $\frac{n}{n+1}$, and the series of reciprocal squares, whose ratio is $\frac{n^2}{(n+1)^2}$. **(a)** Compute both ratio-test limits. **(b)** Using the fact that the two series behave differently, explain why a ratio-test limit of 1 can never be used as evidence either way, and name the test that settles each one. *(Open response — your teacher checks part (b).)*

Answer: ____________________

**6.** A power series is centred at $c = 2$, and a ratio test shows it converges when the distance from the centre is less than the radius $r = 3$. Give the two endpoints of the interval. A student who reports one endpoint as 3 has made a specific mistake — name it.

Answer: ____________________

**7. Find and fix.** A student reports the interval of convergence for that power series as an open interval, on the grounds that "the ratio test gave a strict inequality". Test the left endpoint properly: there the series becomes $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$. Compute its exact sum, and state whether the left endpoint belongs to the interval of convergence. A student who answers $\ln 2$ has dropped a sign — say why the sum must be negative.

Answer: ________________

**8. Challenge.** The right endpoint of the same power series turns the series into the harmonic series. Write the complete interval of convergence, using the correct bracket at each end, and justify each end by naming the test you used there. Then explain, in two or three sentences, why no ratio-test computation can ever decide an endpoint — what is it about the limit at the endpoint that puts it out of reach?